

Part A - Question 1

TP (True Positive) = 140 **FN (False Negative) = 10**

FP (False Positive) = 30 **TN (True Negative) = 20**

Total Patients = 140 + 30 + 10 + 20 = 200

1. Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = \frac{140 + 20}{140 + 30 + 10 + 20} = \frac{160}{200} = 0.80$$

2. Misclassification Rate or Error rate

$$Error\ Rate = \frac{FP + FN}{Total} = \frac{10 + 30}{200} = \frac{40}{200} = 0.20$$

3. Precision or Positive Precision Value (PPV)

$$Precision = \frac{TP}{TP + FN} = \frac{140}{140 + 10} = \frac{140}{150} = 0.933$$

4. Recall or Sensitivity/True Positive Recall (TPR)

$$TPR = \frac{TP}{TP + FN} = \frac{140}{140 + 10} = \frac{140}{150} = 0.933$$

5. Interpretation

The model is 80% accurate in prediction, only 20% of error rate and a 93.3% of precision achieved. The model detects 82.4% of disease cases and only 17.6% of disease go undetected.

Part A - Question 2

TP (True Positive) = 180 **FN (False Negative) = 20**

FP (False Positive) = 30 **TN (True Negative) = 70**

Total Patients = 180+20+30+70 = 300

1. Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = \frac{180 + 70}{300} = \frac{250}{300} = 0.833$$

2. Misclassification Rate or Error rate

$$Error\ Rate = \frac{FP + FN}{Total} = \frac{20 + 30}{300} = \frac{50}{300} = 0.167$$

3. Precision or Positive Precision Value (PPV)

$$Precision = \frac{TP}{TP + FN} = \frac{180}{180 + 30} = \frac{180}{210} = 0.857$$

4. Recall or Sensitivity/True Positive Recall (TPR)

$$TPR = \frac{TP}{TP + FP} = \frac{180}{180 + 20} = \frac{180}{200} = 0.90$$

5. Interpretation

The model is 83.3% accurate in prediction whether the emails are spam or not spam, only 16.7% of error rate and a 85.7% of precision achieved. The model detects 90% of emails are spam and only 10% of them are not spam.

Part A - Question 3

TP (True Positive) = 80 **FN (False Negative) = 10**

FP (False Positive) = 20 **TN (True Negative) = 90**

Total Patients = 80+10+20+90 = 200

1. Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = \frac{80 + 90}{200} = \frac{170}{200} = 0.85$$

2. Misclassification Rate or Error rate

$$Error Rate = \frac{FP + FN}{Total} = \frac{10 + 20}{200} = \frac{30}{200} = 0.15$$

3. Precision or Positive Precision Value (PPV)

$$Precision = \frac{TP}{TP + FN} = \frac{80}{80 + 20} = \frac{80}{100} = 0.80$$

4. Recall or Sensitivity/True Positive Recall (TPR)

$$TPR = \frac{TP}{TP + FP} = \frac{80}{80 + 10} = \frac{80}{90} = 0.889$$

5. Interpretation

The model is 85% accurate in prediction whether the customers churn or loyal, only 15% of error rate and an 80% of precision achieved. The model detects 88.9% of customer churn and only 11.1% of customers under loyal.

Part B

	Item	Description
1	Overfitting	Overfitting occurs when a machine learning model learns the training data too well. This cause the model to perform very well on training data but poorly on new an unseen data because it has failed to generalize.
2	Underfitting	Underfitting happens when a model is too simple to capture the underlying patterns in the data. I perform poorly on both training and testing data.
3	Hold-out Validation	It involves splitting the dataset into two parts: training set and testing set. The model is trained on training set and evaluated on testing set.
4	K-fold Cross Validation	It divides the dataset into K equal parts. The model is trained on K-1 folds and tested on remaining fold.
5	Training set	The training set is the portion of the dataset used to train the model. The model learns patterns, relationships and weights from this data.
6	Testing set	The testing set is the portion of the dataset used to evaluate the trained model. It tests the model's ability to generalize to new data.
7	Target variable/Label	Target variable is the output or result the model is trying to predict.
8	Classification	Classification is a type of supervised learning where the goal is to predict a discrete label or category, such as spam/not spam or pass/fail.
9	Regression	A type of supervised learning where the goal is to predict a continuous value.
10	Stratified sampling	Stratified sampling ensures that each class or category in the dataset is proportionally represented in the sample. This is useful in classification tasks to avoid imbalanced datasets.